Semantic Epistemic Risk versus Confidence-Based Active Retrieval During Generation

Working Draft

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Abstract

Active retrieval during generation is established prior art: FLARE retrieves when forecast tokens have low model confidence [1]. We ask whether uncertainty is equivalent to epistemic risk. Using one versioned controlled source, we freeze 80 untouched Qwen-measured test examples with exactly 20 in each confidence–retrieval-requirement quadrant, then evaluate Qwen3-0.6B, SmolLM2-1.7B, and Gemma3-1B. A transparent semantic trigger has 100% retrieval recall and 2.5% false activation on all three families. Confidence-only recall is 50%, 0%, and 97.5%, respectively, exposing strong checkpoint calibration dependence. A deterministic runtime epistemic gate reduces Unsupported Commitment Rate (UCR) from 100% unassisted to 0% for every model while retrieving less than upfront RAG; accuracy is 97.5%, 96.25%, and 77.5%, so enforcement can trade utility for support. A secondary placement experiment is negative: three rank-8 LoRA adapters achieve near-zero synthetic development loss but collapse to zero held-out retrieval recall. Gemma in-context learning reaches full recall but only 36.25% valid-interface success. Under this freeze, stable retrieval policy belongs in the transparent runtime, not the tested context or weights. The multi-source Paper 2.5 gate passes, but learned source routing is not implied.

1 Question and Claim Boundary

The primary question is: *is epistemic risk distinct from model uncertainty?* Retrieval-required means that a reliable answer depends on an external or source-specific record rather than model weights or already supplied active context. This operational label is distinct from model confidence.

The secondary question is placement: *where should stable one-source interface knowledge live: context, weights, or runtime?* Phase B begins only after the Phase A source, benchmark, confidence probes, threshold objective, and evaluation are frozen. The paper does not claim active retrieval itself as novel and does not study multi-source routing.

2 Mechanism

$$Q \to y_{1:t} \to \widehat{y}_{t+1:t+k} \to \{T_{\text{conf}}, T_{\text{sem}}\} \to \text{retrieve} \to \text{evidence/status} \to \text{accept or abstain.}$$

The FLARE-like controller greedily forecasts one short answer span, records the probability of every forecast token, triggers on the minimum probability, and regenerates with evidence. Canonical FLARE forecasts longer text, may mask uncertain spans, and iterates during long-form generation [1]; our single-opportunity controlled diagnostic is not a canonical reproduction.

The semantic controller uses answer-independent lexical, entity, temporal, freshness, source-dependence, and active-context rules. It has an explicit exception for "electric current" and treats compute requests as belonging to a different coprocessor. Rules never contain source answer values.

3 Controlled Source and Evidence Policy

The frozen source is `atlas-controlled-registry@2026.08.2-frozen-candidates`. Records have stable IDs, validity intervals, source version, and provenance. They cover stable, superseded familiar, unfamiliar private, unavailable, and conflicting facts. Requests contain entity, attribute, as-of date, forecast, source version, and resource bounds. Outcomes include SUPPORTED, CONTRADICTED, UNVERIFIED, STALE, AMBIGUOUS, and CONFLICT.

The runtime gate enforces support relative to this configured source; it does not establish truth. A unique supported value may be accepted. Missing, ambiguous, or conflicting evidence forces abstention rather than an unsupported factual commitment.

4 Frozen Measured-Quadrant Benchmark

We generate 256 candidates: 32 development and 224 candidate-test examples. Confidence is measured on pinned Qwen3-0.6B before test selection. The development threshold maximizes retrieval-requirement F1 minus 0.25 times retrieval rate and equals 0.84993. The candidate pool contains 25, 87, 22, and 90 examples in the four observed Qwen quadrants; the freezer selects exactly 20 from each. IDs are fixed before any condition evaluation.

Retrieval-required subclasses are fresh/current, source-version-specific, private, changed familiar, unavailable, conflicting, exact attribution, and structured one-source values. Retrieval-not-required subclasses are stable facts, supplied context, freshness distractors, quotations, hypotheticals, non-factual prose, and compute needs. The final benchmark contains 32 development and 80 untouched test examples.

Each model independently fits its threshold on the common development set: 0.84993 for Qwen, 0.14693 for SmolLM2, and 0.99971 for Gemma. Qwen and Gemma observe all four test quadrants. SmolLM2 places no test example below its fitted threshold, so its test set lacks low-confidence quadrants. This is a calibration result and limits cross-model quadrant claims.

5 Conditions and Metrics

Phase A has 13 conditions: LLM only, anti-hallucination prompting, upfront RAG, explicit retrieval, FLARE-like confidence, semantic, confidence OR semantic, confidence AND semantic, retrospective verification, evidence advisory, evidence plus abstention instruction, runtime epistemic gate, and oracle.

UCR is the fraction of retrieval-required opportunities that emit an unsupported factual commitment. Authorized Commitment Coverage is the fraction of non-abstention retrieval-required opportunities answered with the authorized value. We also report exact accuracy, retrieval precision/recall and rate, confident-hallucination catch rate, quadrant retrieval, runtime enforcement, generated tokens, calls, and wall time. Accuracy intervals and complete threshold sweeps are machine-readable.

6 Phase A Results

All runs use greedy decoding, seed 17, pinned revisions, PyTorch 2.13.0+XPU, and Intel integrated graphics. Each family produces 1,456 prediction rows and 1,456 traces. Table 1 reports the primary conditions.

Table 1: Untouched 80-example Phase A test. Acc.=exact accuracy, Ret.=retrieval rate, Cov.=Authorized Commitment Coverage.

Model	Condition	Acc.	Ret.	UCR	Cov.
Qwen	LLM only	.238	.000	1.000	.000
	FLARE-like	.625	.500	.675	.325
	Semantic	.613	.513	.250	.750
	Conf. OR semantic	.838	.763	.250	.750
	Runtime gate	.975	.763	.000	1.000
	Upfront RAG	.850	1.000	.250	.750
SmolLM2	LLM only	.463	.000	1.000	.000
	FLARE-like	.463	.000	1.000	.000
	Semantic	.575	.513	.775	.225
	Conf. OR semantic	.575	.513	.775	.225
	Runtime gate	.963	.513	.000	1.000
	Upfront RAG	.400	1.000	.775	.225
Gemma	LLM only	.463	.000	1.000	.000
	FLARE-like	.813	.900	.325	.675
	Semantic	.800	.513	.300	.700
	Conf. OR semantic	.825	.913	.300	.700
	Runtime gate	.775	.913	.000	1.000
	Upfront RAG	.825	1.000	.300	.700

Confidence and semantic risk are not equivalent. Qwen confidence retrieves both low-confidence quadrants, giving 50% recall and 50% false intervention; the semantic rule retrieves all 40 required cases and one of 40 controls. SmolLM2's fitted confidence retrieves nothing. Gemma confidence has 97.5% recall but 82.5% false activation. The same semantic policy has 100% recall and 2.5% false activation on all three because it is independent of checkpoint logits.

The runtime gate eliminates unsupported commitments in every family. It also shows why reliability and answer accuracy must be separated: Qwen and SmolLM2 improve sharply, whereas Gemma drops five accuracy points below upfront RAG. Enforcement prevents unsupported acceptance but cannot guarantee that the base answer or abstention policy maximizes exact-match utility.

7 Phase B: Context, Weights, or Runtime

The adapter dataset contains 160 balanced training rows and 40 development rows with disjoint entity namespaces and values. It excludes every held-out benchmark entity and answer. Targets contain either a four-field `retrieve` block or `NO_RETRIEVAL`; no target contains an answer value. The overlap audit has zero train/development, held-out entity, or held-out value collisions.

Rank-8 adapters train for two epochs on Qwen, SmolLM2, and Gemma. Final development losses are 0.00028, 0.0004, and 0.0001. This apparent fit does not transfer. Table 2 compares held-out policy selection. Interface success additionally requires the correct entity, attribute, date, and source. Prompt-token values are measured condition totals; the runtime semantic rows use the Phase A answer template and are not a matched estimate of policy-only prompt overhead.

All adapters collapse to `NO_RETRIEVAL`: zero recall, no false activation, and 50% balanced selection accuracy. Qwen context behaves the same. SmolLM2 context mostly over-triggers and often emits invalid blocks. Gemma context detects all needs but only 36.25% of all examples pass the complete interface criterion. Adding confidence to adapter outputs does not recover Qwen or SmolLM2 and inherits Gemma confidence's 82.5% false activation.

Near-zero development loss alongside zero held-out recall is direct evidence of protocol over-

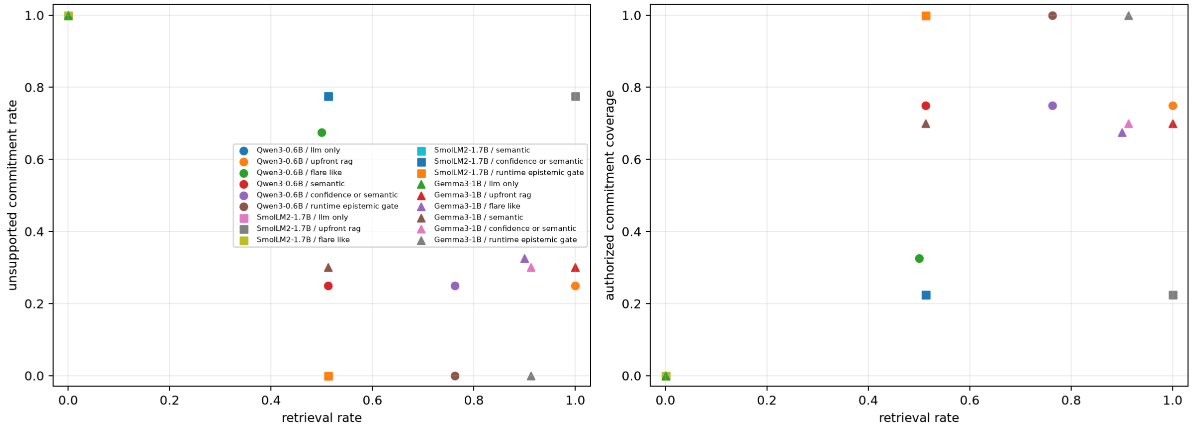

Figure 1: Retrieval cost against UCR and Authorized Commitment Coverage across three checkpoints. The runtime gate reaches zero UCR but has model-dependent accuracy consequences.

Table 2: Held-out 80-example placement evaluation. Sel.=selection accuracy, Int.=valid interface success, Rec.=retrieval recall, FA=false activation.

Model	Placement	Sel.	Int.	Rec.	FA	Prompt tok.	Gen. tok.
Qwen	Context	.500	.500	.000	.000	136.6	6.2
	Weights	.500	.500	.000	.000	67.6	6.0
	Runtime semantic	.988	.988	1.000	.025	101.4	10.2
SmolLM2	Context	.050	.013	.075	.975	172.2	31.4
	Weights	.500	.500	.000	.000	91.2	7.0
	Runtime semantic	.988	.988	1.000	.025	137.0	8.2
Gemma	Context	.863	.363	1.000	.275	139.0	24.4
	Weights	.500	.500	.000	.000	66.0	5.8
	Runtime semantic	.988	.988	1.000	.025	101.1	8.3

fitting to narrow synthetic phrasing. It falsifies the proposed weight-placement benefit under this training design. More diverse training may change that result, but adapting after inspecting these held-out failures would require a new freeze.

8 Paper 2.5 Gate

The pre-registered gate asks whether at least two families reduce UCR with the runtime gate while retrieving less often than upfront RAG. Qwen, SmolLM2, and Gemma all pass; Paper 2.5 is therefore GO. This authorizes testing source heterogeneity. It does not authorize learned source routing, because Phase B provides no evidence that the retrieval policy should move into weights.

9 Falsification and Limitations

The primary result would be weakened if a better-calibrated FLARE implementation matches semantic routing at equal cost, if natural-language semantic rules lose precision, or if the confident-risk quadrants disappear on new data. The present result instead finds checkpoint-dependent confidence behavior and stable rule recall, but only on one controlled source and synthetic short answers.

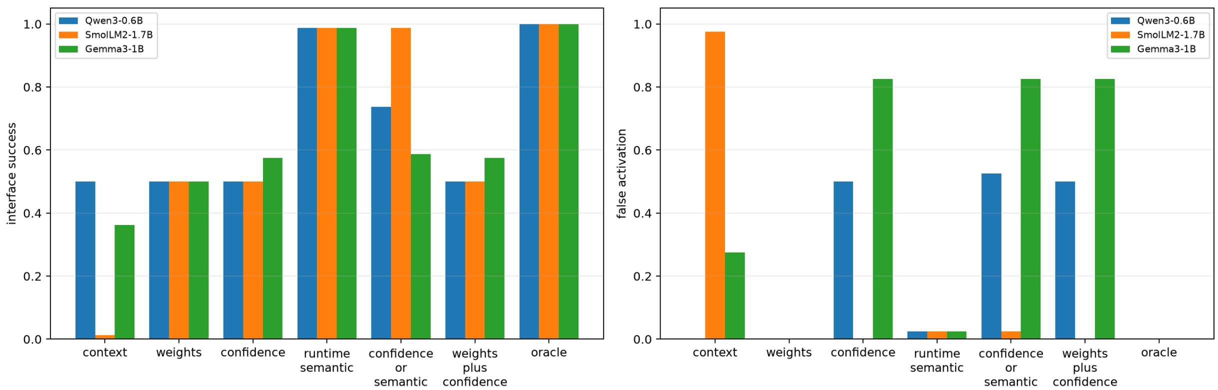

Figure 2: Interface success and false activation for context, weights, confidence, transparent runtime semantics, combinations, and oracle.

The benchmark is selected using Qwen confidence, so its exact 20-per-quadrant balance does not transfer to other checkpoints. SmolLM2 lacks low-confidence test examples. There is one seed per model, no long-form repeated retrieval, and no ranking noise. The semantic cue distribution is controlled. The runtime gate is deterministic and answer-aware relative to source evidence, not a production truth system. Full-prefix generation latency is not production latency.

Phase B uses narrow train templates despite strict entity/value leakage controls; its failure may be covariate shift rather than a general limitation of LoRA. Conversely, development loss is clearly not evidence of interface generalization. The ICL baseline uses textual examples rather than native tool schemas. No condition trains or tests multi-source selection.

10 Reproduction

Paper-specific code is under `src/ccpu/paper1_5`. From the repository root, activate the validated XPU environment and run:

```
$py = 'C:\Users\j.simao\.venvs\modal-llm-xpu\Scripts\python.exe'
$env:PYTHONPATH = (Resolve-Path src).Path
& $py -m ccpu paper1.5 freeze-next \
  --config configs/paper1_5/next_iter_freeze_qwen_xpu.json \
  --output-dir artifacts/paper1_5/next_iter/freeze_qwen_v2
& $py -m ccpu paper1.5 run-hf \
  --config configs/paper1_5/next_iter_qwen_xpu.json \
  --output-dir artifacts/paper1_5/next_iter/qwen_v2
python -m ccpu paper1.5 analyze-next \
  --config configs/paper1_5/next_iter_analysis.json \
  --output-dir artifacts/paper1_5/next_iter/analysis_v1
& $py -m ccpu paper1.5 train-policy-lora <arguments in README>
python -m ccpu paper1.5 analyze-policy \
  --config configs/paper1_5/policy_placement.json \
  --output-dir artifacts/paper1_5/next_iter/policy_analysis_v1
```

Manifests hash configurations, benchmark/source snapshots, predictions, traces, summaries, and adapter files and record checkpoint revisions, device, package, repository, and environment provenance.

11 Conclusion

Semantic epistemic risk adds information beyond checkpoint uncertainty in this controlled one-source regime. A runtime gate eliminates unsupported commitments for three small model families at lower retrieval cost than upfront RAG, although Gemma exposes an accuracy cost. The placement extension is negative: context is inconsistent and the tested adapters overfit. For the current freeze, exact source mechanics and enforcement clearly belong in runtime, and stable semantic selection should remain transparent there until independently frozen training demonstrates weight-policy generalization.

References

[1] Z. Jiang et al. Active retrieval augmented generation. *EMNLP*, 2023. `https://arxiv.org/abs/2305.06983`.